

Experiment - 5

Data Analytics II

1. Implement logistic regression using Python/R to perform classification on Social_Network_Ads.csv dataset.
2. Compute Confusion matrix to find TP, FP, TN, FN, Accuracy, Error rate, Precision, Recall on the given dataset.

```
In [1]: import numpy as np
import matplotlib.pyplot as plt
import pandas as pd
```

```
In [4]: dataset = pd.read_csv(r"D:\College\TE\SEM-2\Practical\DSBDA\5\Social_Network_Ads.csv")
```

```
In [5]: dataset.head()
```

Out[5]:

	User ID	Gender	Age	EstimatedSalary	Purchased
0	15624510	Male	19	19000	0
1	15810944	Male	35	20000	0
2	15668575	Female	26	43000	0
3	15603246	Female	27	57000	0
4	15804002	Male	19	76000	0

```
In [6]: dataset.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 400 entries, 0 to 399
Data columns (total 5 columns):
User ID           400 non-null int64
Gender            400 non-null object
Age              400 non-null int64
EstimatedSalary  400 non-null int64
Purchased        400 non-null int64
dtypes: int64(4), object(1)
memory usage: 15.8+ KB
```

```
In [7]: dataset.isnull().sum()
```

```
Out[7]: User ID           0
Gender            0
Age              0
EstimatedSalary  0
Purchased        0
dtype: int64
```

```
In [8]: dataset.shape
```

```
Out[8]: (400, 5)
```

```
In [10]: x = dataset.iloc[:, [2, 3]].values
y = dataset.iloc[:, 4].values
```

```
In [11]: print(x)
```

```
[[ 19 19000]
 [ 35 20000]
 [ 26 43000]
 [ 27 57000]
 [ 19 76000]
 [ 27 58000]
 [ 27 84000]
 [ 32 150000]
 [ 25 33000]
 [ 35 65000]
 [ 26 80000]
 [ 26 52000]
 [ 20 86000]
 [ 32 18000]
 [ 18 82000]
 [ 29 80000]
 [ 47 25000]
 [ 45 26000]
 [ 46 28000]
```

```
In [12]: from sklearn.model_selection import train_test_split
x_train, x_test, y_train, y_test = train_test_split(x, y, test_size = 0.30, random_stat
```

```
In [13]: from sklearn.preprocessing import StandardScaler
sc = StandardScaler()
x_train = sc.fit_transform(x_train)
x_test = sc.transform(x_test)
```

```
In [14]: from sklearn.linear_model import LogisticRegression
classifier = LogisticRegression(random_state = 0)
classifier.fit(x_train, y_train)
```

C:\Users\HP\Anaconda3\lib\site-packages\sklearn\linear_model\logistic.py:432: FutureWarning: Default solver will be changed to 'lbfgs' in 0.22. Specify a solver to silence this warning.
FutureWarning)

```
Out[14]: LogisticRegression(C=1.0, class_weight=None, dual=False, fit_intercept=True,
intercept_scaling=1, l1_ratio=None, max_iter=100,
multi_class='warn', n_jobs=None, penalty='l2',
random_state=0, solver='warn', tol=0.0001, verbose=0,
warm_start=False)
```

```
In [15]: y_pred = classifier.predict(x_test)
```

```
In [16]: print(y_pred)
```

```
[0 0 0 0 0 0 0 1 0 1 0 0 0 0 0 0 0 0 1 0 0 1 0 1 0 1 0 0 0 0 0 0 1 0 0 0 0
 0 0 1 0 0 0 0 1 0 0 1 0 1 1 0 0 0 1 0 0 0 1 0 0 1 0 0 0 1 0 0 0 0 1 0 0 0
 0 0 1 0 1 1 1 0 0 1 1 0 1 0 0 0 1 0 0 0 0 0 0 1 1 0 0 0 0 1 1 0 0 0 0 0
 0 0 1 1 1 1 0 1 1]
```

```
In [17]: print(y_test)
```

```
[0 0 0 0 0 0 0 1 0 0 0 0 0 0 0 0 0 1 0 0 1 0 1 0 1 0 0 0 0 0 1 1 0 0 0 0
 0 0 1 0 0 0 0 1 0 0 1 0 1 1 0 0 0 1 1 0 0 1 0 0 1 0 1 0 1 0 0 0 0 1 0 0 1
 0 0 0 0 1 1 1 0 0 0 1 1 0 1 1 0 0 1 0 0 0 1 0 1 1 1 0 1 1 1 0 1 1 0 0 0 0 0
 0 1 1 1 0 1 0 0 1]
```

```
In [18]: from sklearn.metrics import confusion_matrix
cm = confusion_matrix(y_test, y_pred)
print(cm)
```

```
[[74  5]
 [10 31]]
```

```
In [19]: Accuracy=(74+31)/120
Accuracy
```

```
Out[19]: 0.875
```

```
In [20]: Error_rate=(5+10)/120
Error_rate
```

```
Out[20]: 0.125
```

```
In [21]: from sklearn.metrics import precision_score, recall_score
precision_score(y_test, y_pred)
```

```
Out[21]: 0.8611111111111112
```

```
In [22]: recall_score(y_test, y_pred)
```

```
Out[22]: 0.7560975609756098
```

```
In [23]: from sklearn.metrics import f1_score
f1_score(y_test, y_pred)
```

```
Out[23]: 0.8051948051948052
```